

Highlights

Stochastic Physical-Ensemble Quantiles Improve Hybrid Machine-Learning Simulation of Daily Streamflow in the High-Andean Ramis Basin

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- Ensemble quantiles were the dominant driver of hybrid streamflow skill.
- The quantile hybrid improved NSE by 0.311 over stochastic RAMIS.
- Lévy perturbations informed ML features but underdispersed intervals.
- Backend and leakage audits constrained claims for GRU-labelled runs.
- The ablation separated memory, stochasticity, ML and descriptors.

Stochastic Physical-Ensemble Quantiles Improve Hybrid Machine-Learning Simulation of Daily Streamflow in the High-Andean Ramis Basin

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ABSTRACT

Hybrid hydrological modelling combines process-based structure with the predictive flexibility of machine learning (ML), but often leaves unclear which component drives improved streamflow prediction. We present StoHyMoLAP, a reproducible and audit-controlled framework for daily streamflow simulation in the high-Andean Ramis basin, within the Lake Titicaca system of southern Peru. The framework couples a lumped RAMIS rainfall-runoff core, an optional linear baseflow reservoir, Lévy-type (α -stable) stochastic perturbations, Monte Carlo physical ensembles, and ML emulators trained from observed forcings or from stochastic-ensemble descriptors. Nine experiments (E0–E8) were evaluated over a common validation window from 8 January 2009 to 31 December 2020 ($n = 4367$ daily samples), with temporal-leakage and backend audits embedded in the evidence chain.

Quantile descriptors from the stochastic physical ensemble provided the largest robust gain. The gradient-boosting quantile hybrid reached NSE = 0.740 and KGE = 0.810, improving NSE by 0.311 relative to the complete stochastic physical model and by 0.165 relative to the pure ML baseline. Replacing ensemble-mean features with ensemble quantiles alone added 0.114 NSE, showing that distributional physical descriptors contain information not captured by a single mean trajectory. The baseflow reservoir modestly improved the deterministic physical baseline, whereas direct Lévy perturbations produced inconsistent point-prediction gains. The stochastic physical bands were strongly underdispersive (PICP = 0.004 against a nominal 0.95 target), so the ensemble is more useful as ML features than as calibrated predictive uncertainty. The numerically best run (NSE = 0.849, KGE = 0.899, RMSE = $0.210 \text{ m}^3 \text{ s}^{-1}$) was labelled as GRU but executed as a fallback multilayer perceptron on flattened sequences; it is not evidence of recurrent-network superiority. These findings support stochastic physical-ML hybridization while showing that uncertainty calibration, low-flow correction and backend-aware reporting remain essential for defensible hydrological modelling.

1. Introduction

Daily streamflow simulation in ungauged or poorly instrumented basins remains one of the most practically consequential open problems in surface water hydrology (?). The challenge is simultaneously structural and statistical: process-based models impose a parsimonious prior on catchment behaviour but are limited by parameter uncertainty, simplified storage representations and, in high-altitude settings, by severe scarcity of ground observations (??). Pure machine-learning (ML) models can exploit nonlinear empirical relations and have repeatedly matched or exceeded the performance of calibrated conceptual models for streamflow prediction at the basin scale (???), yet their predictions depend entirely on the statistical distribution of training data and can be fragile when that distribution shifts (??). A further structural limitation is that “ML models have almost no process understanding capabilities like process-based models beyond their fitting capability” (?), which translates into poor reliability at flow extremes and under non-stationary climate forcing (?).

Hybrid approaches address these complementary weaknesses by feeding physically simulated variables as structured predictors into a data-driven correction model, thereby combining process knowledge with empirical flexibility (??). The literature on such hybridization has grown rapidly. ? showed that a SWAT-ANN hybrid outperforms the individual models across calibration, validation and test periods, with especially pronounced gains at low and high

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flows. ? demonstrated that “nested” hybridization, where all parameters are optimized jointly, outperforms parallel coupling in terms of both NSE and RMSE reduction ($\sim 24\text{--}30\%$ over the standalone conceptual model). ? showed that coupling IHACRES with an MLP improved RMSE by approximately 27%, concluding that the approach “overcomes the basic deficiencies of the conceptual and ML methods.” More recently, ? embedded physically derived baseflow sequences within an LSTM architecture, finding that “the synergy between physics and DL within a unified architecture has yielded an effect where the combination is greater than the sum of its parts.”

A common limitation of these hybridizations is that they are built on *deterministic* hydrological cores, passing forward a single simulated trajectory and discarding information about the conditional spread and tail behaviour of the simulated response. This matters because rainfall–runoff transformation is inherently stochastic: “the spatial and temporal variability of rainfall is responsible for a dispersion of the peak discharge of 20–25%” even when volumetric totals are well specified (?), and the autocorrelation structure of the rainfall field has been shown to systematically bias peak discharge estimates (??). A further diagnostic concern is that conceptual hydrological models “perform poorly when there are large climatic differences between the calibration and validation periods, in catchments with flashy flows, and in catchments with unexplained variations in low flow measurements” (?), which describes precisely the hydrological setting of the high-Andean Altiplano.

The recent study by ? extended hybrid modelling to a *stochastic* physically-based core by integrating statistical properties—mean and quantiles—of the daily discharge probability distribution simulated by a Lévy-driven stochastic HyMoLAP model as inputs to wavelet-based XGBoost and GRU emulators in the Ouémé basin (Benin). Their central finding was that injecting these distributional descriptors improved point discharge prediction, particularly at low and high flows. That work establishes the *feasibility* of stochastic descriptor hybridization, but three questions remain practically open: (i) *which* part of the pipeline produces the gain—hydrological memory, the stochastic noise itself, the ML correction, or the choice of descriptor; (ii) whether the simulated stochastic bands constitute *calibrated* uncertainty intervals or merely informative summaries; and (iii) whether reported architecture labels reflect the backend that actually ran. Resolving these questions requires a controlled ablation and an explicit reproducibility audit (?) rather than a single best-model comparison.

This manuscript presents StoHyMoLAP, a modular platform designed to answer those questions for the Ramis basin. The central research question is whether hydrological memory, Lévy-type stochastic perturbations and physical-ensemble descriptors each improve daily streamflow simulation relative to physical-only and ML-only baselines, and which of them matters most. Our contribution is therefore not a new hybridization mechanism but a *controlled, audited decomposition* of one: nine experiments (E0–E8; Table 1, workflow in Fig. 1) are executed through a single code base under a common-window protocol, and the incremental effect of each component is read off as an ablation contrast. To our knowledge, this is the first application of stochastic physical–ML quantile hybridization to a high-Andean Altiplano catchment, a hydrological setting that differs from the West-African basin of ? in its strong seasonality, semi-arid character and data scarcity above 3 800 m a.s.l.

A critical reproducibility point, stated up front, is that the experiments labelled as GRU in the current run were not executed with a true recurrent backend. The recorded backend for E7 and E8 is `fallback_mlp_on_flattened_sequences`; therefore, these experiments are discussed as GRU-labelled sequential hybrids implemented through a fallback multilayer perceptron (MLP), not as evidence of recurrent neural-network superiority (?).

2. Materials and methods

2.1. Study area

The Ramis River is the largest tributary of Lake Titicaca and the lake’s principal surface inflow, draining roughly one third of the Titicaca catchment in the Puno region of the Peruvian Altiplano (??). The broader Titicaca system receives approximately 53% of its annual water budget from riverine inflow, with evaporation accounting for 91% of losses (?). Observed lake levels have fluctuated by approximately 6 m over the past century, driven primarily by interannual variability in precipitation rather than by snowmelt, irrigation withdrawals or ice melt (??). Simple mass-balance models for the system reproduce the observed seasonal asymmetry—peak levels in April, minimum in December—with amplitudes close to 0.74 m (??), providing a useful physical reference for the catchment-scale discharge dynamics studied here.

The Ramis basin lies in a high-altitude (above $\sim 3\,800$ m a.s.l.), semi-arid Andean environment with a strongly unimodal seasonal precipitation regime: a wet season concentrated between December and March (approximately 60% of annual totals in three months) and a pronounced dry season from roughly April to November (?). The

Altiplano climate is characterized by significant interannual variability (?), and the precipitation gradient from north to south is marked, ranging from about 750 mm yr⁻¹ in the north to well below 200 mm yr⁻¹ in the south (?). This seasonality produces a markedly intermittent, “flashy” hydrological response, with low and sustained dry-season baseflow punctuated by sharp wet-season peaks—a regime known to challenge both deterministic conceptual models (??) and purely data-driven models (?). Conceptual models calibrated for the region have shown that the streamflow response is dominated by groundwater storage fluxes rather than direct routing of snowmelt, underscoring the importance of baseflow representation (?). Daily discharge is gauged at the Puente Ramis station (SENAMHI-Perú).

The gauged drainage area is approximately [FILL: XX XXX] km², the station is located at [FILL: lat, lon], and the daily series covers [FILL: record period].

2.2. Study data and temporal protocol

The modelling unit is a daily hydro-meteorological series with precipitation (P_t), minimum and maximum temperature ($T_{\min,t}$, $T_{\max,t}$), potential evapotranspiration (PET_t) and observed streamflow (Q_t^{obs}). Effective precipitation is

$$P_t^{eff} = \max(P_t - PET_t, 0). \quad (1)$$

All models are trained and validated under a strict chronological split: calibration from 8 September 1991 to 31 December 2008, validation from 8 January 2009 to 31 December 2020. The final comparison is restricted to the common validation intersection of all experiments, comprising $n = 4367$ daily samples. This common-window design avoids the well-documented problem of giving physical models more validation days than ML or sequential models (?), and ensures that all performance contrasts are computed on an identical set of observations.

The temporal protocol embodies the leakage-control principles recommended by ? and ?: predictors are constructed with non-negative lags only, sequence samples are generated within each temporal subset, scalars are fitted exclusively on training data, and flow-regime thresholds are estimated from training observations alone. The full audit is summarized in Table 7. Partitioning meteorological inputs and observational scalars by temporal subset is particularly important when stochastic ensemble summaries—which aggregate distributional properties of the simulated discharge—are used as ML features: fitting their normalization on the full sample would constitute a subtle form of the temporal leakage documented by ?.

2.3. Physical rainfall–runoff core

Despite the term “hydrodynamic” sometimes appearing in the user-facing workflow, the current StoHyMoLAP implementation is a lumped stochastic rainfall–runoff simulator rather than a spatial two-dimensional solver. The physical core is therefore described here as a discharge-dynamics model. Its fast-flow component follows a RAMIS-type state recurrence:

$$x_t = \left(1 - \frac{\mu}{\lambda}\right) x_{t-1} + P_t^{eff}, \quad (2)$$

where $\mu > 0$ and $\lambda > 0$ control memory and recession stability. The constraint $0 < \mu/\lambda < 1$ prevents unstable or sign-changing memory factors. The fast discharge trajectory Q_t^{fast} is

$$Q_t^{fast} = Q_{t-1}^{fast} - \frac{\mu}{\lambda} Q_0 \left(\frac{\max(Q_{t-1}^{fast}, 0)}{Q_0} \right)^{2\mu-1} + \frac{\alpha_A}{\lambda} x_{t-1} + \sigma Q_{t-1}^{fast} \Delta L_t, \quad (3)$$

where Q_0 is the reference discharge, α_A is an empirical conversion factor, σ controls stochastic amplitude, and ΔL_t is a Lévy-stable increment. Negative simulated discharge is clipped to zero when the experiment activates non-negative physical post-processing.

In line with ?, the recurrence in Eqs. 2–3 treats catchment-scale nonlinearity as emerging from the aggregation of heterogeneous storage elements rather than from local hydraulic nonlinearity. This allows the parameter μ to be interpreted as a shape parameter that implicitly encodes the heterogeneity of the storage ensemble, consistent with the recession analysis framework of that work.

2.4. Baseflow reservoir as hydrological memory

Hydrological memory is represented by an optional linear baseflow reservoir. The theoretical basis for this component rests on the observation that in semi-arid Andean catchments like the Ramis, “the streamflow response is mostly consistent with flux from groundwater storage” (?), and that models ignoring this dynamics tend to overestimate seasonal low-flow periods and underestimate high-flow periods (?). Furthermore, ? demonstrated quantitatively that including previous runoff (a proxy for storage memory) as a model input “considerably improves the accuracy [with] increments in RMSE and MAE generally more than 90%,” and that “excluding previous runoff information from the model inputs or including only rainfall information produces inaccurate models.”

Recharge to the store is proportional to effective precipitation:

$$R_t = c_r \alpha_A P_t^{eff}, \quad (4)$$

where $c_r \in [0, 1]$ is a recharge coefficient. Storage and baseflow discharge evolve as

$$S_{b,t+1} = \max(0, S_{b,t} + R_t - k_b S_{b,t}), \quad (5)$$

$$Q_t^{base} = k_b S_{b,t}, \quad (6)$$

with $k_b \in [0.001, 0.5]$, and the total physical discharge is

$$Q_t^{phys} = \max(Q_t^{fast} + Q_t^{base}, 0). \quad (7)$$

This component is activated in E1, E3 and the hybrid configurations. Its isolated effect is quantified through the E1–E0 and E3–E2 ablation contrasts in Table 3.

2.5. Lévy stochastic perturbations and Monte Carlo ensemble

The stochastic physical experiments inject α -stable Lévy increments into Eq. 3. Random variates are generated with the Chambers–Mallows–Stuck method (?), parameterized by stability α , skewness β , location and scale. The use of heavy-tailed α -stable perturbations rather than Gaussian noise is motivated by the documented importance of higher moments in hydrological ensemble prediction: ? noted that “biases occur not only in relation to the mean, but also in higher statistical moments of the predictive distribution,” and ? showed empirically that “poorer sampling of the higher moments of the distribution” limits skill when ensemble members are insufficient to characterize the tail. In the strongly seasonal, intermittent Ramis regime, where the distribution of daily discharge is far from Gaussian, heavy-tailed perturbations provide a more conservative and physically plausible stochastic forcing than Gaussian equivalents.

For each calibrated parameter set, a Monte Carlo ensemble of N trajectories is simulated, following the two-step design—stochastic input generation followed by hydrological simulation—recommended by ? and ?. The ensemble is summarized at each time step by the mean, median, standard deviation and quantiles

$$\{q_{0.01}, q_{0.05}, q_{0.25}, q_{0.50}, q_{0.75}, q_{0.95}, q_{0.99}\}, \quad (8)$$

as well as interquantile widths such as $q_{0.95} - q_{0.05}$ and $q_{0.75} - q_{0.25}$. The use of quantile-based representations is motivated by their robustness to fat-tailed distributions and outliers (?), which makes them suitable summaries of an α -stable ensemble even when the distribution cannot be characterized by its first two moments alone. These descriptors serve two purposes: direct stochastic physical simulation in E2 and E3, and physically informed features for hybrid emulators in E5–E8. The uncertainty diagnostics derived from these ensembles are reported in Table 5 and Fig. 7.

2.6. Machine-learning emulator architecture

The ML component is a supervised emulator mapping feature vectors $\mathbf{z}_t$ to simulated discharge:

$$\hat{Q}_t = f_\theta(\mathbf{z}_t), \quad (9)$$

followed by non-negative clipping:

$$Q_t^{sim} = \max(\hat{Q}_t, 0). \quad (10)$$

Three feature families are used. First, the pure ML baseline (E4) uses observed forcing variables and their non-negative lags. The inclusion of lagged observed discharge as a feature is consistent with the strong empirical evidence

that lagged runoff markedly improves ML-based rainfall–runoff models, and that pure meteorological inputs alone are “inappropriate for hourly runoff prediction” (?). Second, mean-based hybrids (E5, E7) use the ensemble mean of the stochastic physical simulation, arranged as temporal sequences for the sequential emulator. Third, quantile-based hybrids (E6, E8) use ensemble quantiles and spread descriptors, preserving information about the shape and tail behaviour of the physical ensemble. The theoretical justification for the move from mean to quantile descriptors rests on the argument of ? that “there are two ways for improving predictive performance: (i) improving the prediction algorithm; and (ii) discovering new informative predictors,” and that quantile representations of ensemble distributions constitute precisely the kind of additional informative predictor that enables probabilistic post-processing to go beyond first-moment correction.

The tabular emulator is labelled as XGB. If the `xgboost` package is available, the implementation uses `XGBRegressor` (?); otherwise it falls back to `HistGradientBoostingRegressor` (?). In the current evidence package, the XGB-labelled models used the `sklearn_hgb` backend (Table 6); both are gradient-boosted tree ensembles, so the XGB-labelled results remain valid evidence of gradient-boosting hybridization. Hyperparameter search uses randomized search with `TimeSeriesSplit`, preserving temporal ordering during model selection (?).

The sequential emulator is labelled as GRU (?) and can use TensorFlow/Keras or PyTorch GRU layers when available. If neither backend is installed, the implementation flattens each sequence and trains an `MLPRegressor`. The current E7 and E8 runs used this fallback. Therefore, in all Results and Discussion statements, E7 and E8 are interpreted as GRU-labelled sequential hybrids with fallback MLP backends, not as true recurrent GRU models.

2.7. Calibration objective

The physical and stochastic models are calibrated with a scalar multi-objective score combining efficiency, distributional agreement, bias and interval coverage:

$$J = \frac{w_{NSE}(1 - NSE) + w_{KGE}(1 - KGE) + w_{PBIAS} \left| \frac{PBIAS}{100} \right| + w_{cov}|c^* - PICP|}{w_{active}}, \quad (11)$$

where $c^* = 0.95$ is the target coverage and w_{active} is the sum of active weights. The default weights are $w_{NSE} = 0.40$, $w_{KGE} = 0.30$, $w_{PBIAS} = 0.15$ and $w_{cov} = 0.15$. When a model provides no prediction intervals, the coverage term is omitted and remaining weights are renormalized.

This composite criterion follows the recommendation of ? that “there is no single criterion that can evaluate model performance in all cases,” and the argument of ? that “scoring rules that evaluate either reliability or sharpness alone are not proper for the task.” The inclusion of KGE as a complementary metric to NSE is motivated by its well-known decomposition into correlation, variability and bias components (?), which makes it less sensitive than NSE to the systematic underweighting of low-flow errors discussed by ?. The coverage term penalizes underdispersive stochastic bands (??).

2.8. Experimental matrix and ablation design

The experimental matrix in Table 1 isolates the contribution of each modelling component. E0 and E1 test deterministic RAMIS without and with baseflow. E2 and E3 test Lévy stochasticity without and with baseflow. E4 is a pure ML baseline. E5 and E6 are XGB-labelled hybrids using mean and quantile physical-ensemble features, respectively. E7 and E8 extend these to sequential emulators with the fallback MLP backend in the current run.

2.9. Relation to prior stochastic physical–ML hybridization

The idea of feeding statistical descriptors of a stochastic physical ensemble to an ML emulator follows ?. The present study is complementary rather than overlapping. ? *demonstrate* that stochastic descriptor hybridization improves point prediction in a single basin and report a best hybrid model. The present study instead *decomposes* the same family of pipelines: (i) it isolates the marginal contribution of hydrological memory, stochastic noise, ML correction and descriptor choice through nine controlled ablation contrasts on a common validation window; (ii) it adds a linear baseflow reservoir as an explicit memory component and evaluates it in isolation; (iii) it reports KGE and PBIAS alongside NSE/RMSE and adds flow-regime error analysis; (iv) it evaluates the stochastic bands as *calibrated* uncertainty using coverage diagnostics (PICP, PINAW, Winkler) following the probabilistic verification framework of ? and ?; and (v) it embeds a leakage-and-backend audit, including the disclosure that the GRU-labelled runs used a fallback MLP (?). The application setting also differs markedly: a high-Andean, strongly seasonal Altiplano basin versus a West-African basin.

Table 1

Controlled experimental matrix. Backend entries report the actual backend used in the current run.

Experiment	Family	Lévy	Baseflow	Emulator	Features	Backend used
E0_RAMIS_DET_NOBF	physical RAMIS	no	no	none	physical_only	physical
E1_RAMIS_DET_BF	physical RAMIS	no	yes	none	physical_only	physical
E2_RAMIS_LEVY_NOBF	RAMIS-Lévy	yes	no	none	physical_only	physical
E3_RAMIS_LEVY_BF	RAMIS-Lévy	yes	yes	none	physical_only	physical
E4_ML_PURE_XGB	pure ML	no	no	XGB-labelled	observed_forcing	sklearn_hgb
E5_HYB_XGB_MEAN	hybrid	yes	yes	XGB-labelled	ensemble mean	sklearn_hgb
E6_HYB_XGB_QUANTILES	hybrid	yes	yes	XGB-labelled	ensemble quantiles	sklearn_hgb
E7_HYB_GRU_MEAN	sequential hybrid	yes	yes	GRU-labelled	mean sequence	fallback MLP
E8_HYB_GRU_QUANTILES	sequential hybrid	yes	yes	GRU-labelled	quantile sequence	fallback MLP

Table 2

Final common-window ranking. Metrics are computed on 4367 validation samples from 8 January 2009 to 31 December 2020.

Rank	Experiment	NSE	KGE	RMSE	MAE	PBIAS	Backend
1	E8_HYB_GRU_QUANTILES	0.849	0.899	0.210	0.124	1.579	fallback_mlp_on_flattened_sequences
2	E6_HYB_XGB_QUANTILES	0.740	0.810	0.275	0.162	0.927	sklearn_hgb
3	E7_HYB_GRU_MEAN	0.686	0.789	0.302	0.192	1.644	fallback_mlp_on_flattened_sequences
4	E5_HYB_XGB_MEAN	0.626	0.715	0.330	0.210	2.007	sklearn_hgb
5	E4_ML_PURE_XGB	0.575	0.661	0.352	0.223	0.742	sklearn_hgb
6	E1_RAMIS_DET_BF	0.438	0.714	0.405	0.248	0.860	physical
7	E3_RAMIS_LEVY_BF	0.429	0.725	0.408	0.255	-1.218	physical
8	E2_RAMIS_LEVY_NOBF	0.427	0.717	0.409	0.268	-7.551	physical
9	E0_RAMIS_DET_NOBF	0.414	0.713	0.413	0.270	-5.865	physical

2.10. Evaluation metrics and audits

Model performance is evaluated with NSE (?), KGE (?), RMSE, MAE, PBIAS and R^2 . Performance is interpreted following ?. Regime metrics are computed for low, mid and high flows using thresholds estimated from calibration observations only; NSE is interpreted with caution for low flows, where observed variance is small (?). Stochastic reference bands are evaluated with PICP, PINAW, mean prediction interval width (MPIW) and Winkler score (??). An ensemble is considered well-calibrated when PICP approaches the nominal level and PINAW is minimized subject to that coverage constraint (??). The final audit checks temporal split integrity, feature construction, target leakage, backend metadata, common-window consistency and non-negative ML post-processing (Table 7).

3. Results

3.1. Common-window ranking

All experiments were evaluated over the common validation window described in Section 2.2. Table 2 and Fig. 2 summarize the final ranking. The strongest *robust* configuration was E6_HYB_XGB_QUANTILES, with NSE = 0.740 and KGE = 0.810: it is the best model whose reported architecture matches the backend that actually ran, and it provides the most defensible evidence for quantile-based physical-ML hybridization. The top-ranked configuration overall was E8_HYB_GRU_QUANTILES, with NSE = 0.849, KGE = 0.899, RMSE = 0.210 m³ s⁻¹, MAE = 0.124 m³ s⁻¹ and PBIAS = 1.579%; however, this result must be read under the backend caveat in Table 6 because E8 was executed as a fallback MLP on flattened sequences. All four hybrid models outperformed the pure ML baseline (E4, NSE = 0.575), which in turn outperformed all physical baselines in NSE and RMSE, consistent with the pattern reported by ? and ? for LSTM-based models in ungauged catchments.

3.2. Hydrograph and flow-duration behaviour

The validation hydrograph in Fig. 3 compares observed and simulated daily discharge for the representative models. Hybrid models better track temporal variability than the physical-only baselines, while the flow-duration curves in

Table 3

Ablation effects on the common validation window. Positive Δ NSE and Δ KGE indicate improvement; negative Δ RMSE and Δ MAE indicate error reduction.

Ablation	Baseline	Candidate	Δ NSE	Δ KGE	Δ RMSE
deterministic_baseflow	E0_RAMIS_DET_NOBF	E1_RAMIS_DET_BF	0.024	0.000	-0.009
stochastic_baseflow	E2_RAMIS_LEVY_NOBF	E3_RAMIS_LEVY_BF	0.002	0.008	-0.009
levy_given_baseflow	E1_RAMIS_DET_BF	E3_RAMIS_LEVY_BF	-0.010	0.011	0.004
pure_ml_vs_physical	E3_RAMIS_LEVY_BF	E4_ML_PURE_XGB	0.147	-0.064	-0.056
hybrid_mean_vs_physical	E3_RAMIS_LEVY_BF	E5_HYB_XGB_MEAN	0.198	-0.010	-0.078
hybrid_quantiles_vs_physical	E3_RAMIS_LEVY_BF	E6_HYB_XGB_QUANTILES	0.311	0.085	-0.133
quantiles_vs_mean_xgb	E5_HYB_XGB_MEAN	E6_HYB_XGB_QUANTILES	0.114	0.095	-0.053
gru_mean_vs_xgb_mean	E5_HYB_XGB_MEAN	E7_HYB_GRU_MEAN	0.060	0.074	-0.028
gru_quantiles_vs_xgb_quantiles	E6_HYB_XGB_QUANTILES	E8_HYB_GRU_QUANTILES	0.109	0.089	-0.060

Fig. 4 show that quantile-based hybridization improves distributional agreement across a broad range of exceedance probabilities. These visual diagnostics are consistent with the metric improvements in Table 2, but they also reveal persistent discrepancies during low-flow periods. The pronounced collapse of the E0 physical model in the high-exceedance tail of the FDC is particularly informative: it reflects the inability of a single-reservoir deterministic model to maintain positive discharge during dry-season recession, a limitation documented for similar semi-arid Andean settings by ? and ?.

3.3. Ablation effects

The ablation contrasts in Table 3 and Fig. 5 quantify the incremental contribution of each modelling component. Adding the baseflow reservoir to the deterministic physical model (E1–E0) increased NSE by 0.024 and reduced RMSE by $0.009 \text{ m}^3 \text{ s}^{-1}$. Under stochastic forcing, the baseflow effect was smaller but remained positive in NSE and KGE (E3–E2). The direct addition of Lévy perturbations under active baseflow (E3–E1) had mixed effects: NSE decreased by 0.010 while KGE improved by 0.011, confirming that Lévy noise does not consistently improve point predictions.

The largest robust improvement came from hybridization. Replacing the physical mean output with the quantile-based hybrid (E6–E3) increased NSE by 0.311 and reduced RMSE by $0.133 \text{ m}^3 \text{ s}^{-1}$, a gain comparable to the 23–27% NSE improvements reported by ? when ML surrogates were enriched with synthetic physical data. The quantile-versus-mean contrast (E6–E5) added 0.114 NSE and 0.095 KGE, isolating the information value of distributional descriptors beyond the ensemble mean alone.

3.4. Performance by flow regime

Regime-specific results (Table 4, Fig. 6) show that performance is strongly flow-dependent. Low-flow NSE values were negative for all selected models, which is expected when the observed variance within the regime is small (?). This reflects a known structural limitation: the NSE criterion is dominated by variance at the highest flows, making it an unreliable indicator of skill in the dry season (??). However, absolute-error metrics improved markedly with hybridization: low-flow RMSE decreased from $0.177 \text{ m}^3 \text{ s}^{-1}$ (E0) to $0.075 \text{ m}^3 \text{ s}^{-1}$ (E6) and $0.047 \text{ m}^3 \text{ s}^{-1}$ (E8). The improvement in low flows from the quantile hybrid is consistent with the finding of ? that LSTM-based models “show a large performance improvement for low-flow bias score (%BiasFLV)” relative to conceptual models, and with ? who found that physically informed architectures significantly enhance low-flow simulations. For high flows, E8 reached the best regime RMSE among the selected models; the remaining gap at high flows— $0.352 \text{ m}^3 \text{ s}^{-1}$ for E8 versus $0.615 \text{ m}^3 \text{ s}^{-1}$ for E0—reflects both the difficulty of extreme events and the benefit of ensemble quantile features in capturing the asymmetry of the upper tail.

3.5. Uncertainty diagnostics and backend audit

The stochastic physical bands were strongly underdispersive (Table 5, Fig. 7). E2 achieved PICP = 0.160 and E3 achieved PICP = 0.004, far below the nominal 0.95 target, with correspondingly high Winkler scores. These results confirm what ? described as the predominant failure mode of hydrological ensemble prediction systems: “most hydrological ensemble prediction systems fail to account for all sources of uncertainty, leading to under-dispersed

Table 4

RMSE ($\text{m}^3 \text{s}^{-1}$) by flow regime for representative models. Regime thresholds were estimated from calibration observations only.

Experiment	Low	Mid	High
E0_RAMIS_DET_NOBF	0.177	0.392	0.615
E1_RAMIS_DET_BF	0.197	0.365	0.613
E3_RAMIS_LEVY_BF	0.185	0.387	0.604
E4_ML_PURE_XGB	0.189	0.276	0.566
E6_HYB_XGB_QUANTILES	0.075	0.222	0.461
E8_HYB_GRU_QUANTILES	0.047	0.171	0.352

Table 5

Uncertainty diagnostics for stochastic physical reference bands.

Experiment	PICP	PINAW	MPIW	Winkler
E2_RAMIS_LEVY_NOBF	0.160	0.084	0.219	7.297
E3_RAMIS_LEVY_BF	0.004	0.003	0.009	10.050

Table 6

Backend-aware manuscript notes for ML and hybrid configurations.

Experiment	ML label	Backend label	Feature source	Manuscript claim
E4_ML_PURE_XGB	xgboost	sklearn_hgb	observed_forcing	report as gradient
E6_HYB_XGB_QUANTILES	xgboost	sklearn_hgb	stochastic_ensemble	report as gradient
E7_HYB_GRU_MEAN	gru	fallback_mlp_on_flattened_sequences	stochastic_ensemble	fallback MLP, not
E8_HYB_GRU_QUANTILES	gru	fallback_mlp_on_flattened_sequences	stochastic_ensemble	fallback MLP, not

Table 7

Reproducibility checklist for the current evidence package.

Item	Status	Evidence
Common validation intersection	PASS	leaderboard_common_intersection.csv
Anti-leakage audit	WARN	leakage_audit_status.txt=WARN
Physical audit	WARN	physical_audit_status.txt=WARN
Figure manifest	PASS	outputs/figures/phase34/phase34_figure_manifest.csv
GRU wording	WARN	E7/E8 use fallback backend
Optional physical components in figures	WARN	skipped entries in phase34_figure_manifest.csv

forecasts.” The stochastic bands should therefore be interpreted as physically structured stochastic references rather than calibrated predictive intervals (??).

Backend metadata are summarized in Table 6. The XGB-labelled experiments (E4–E6) used `sklearn_hgb`, and the GRU-labelled experiments (E7, E8) used `fallback_mlp_on_flattened_sequences`. The reproducibility checklist (Table 7) returned WARN for anti-leakage and physical audits but no FAIL. WARN items were associated with missing observed-flow values, low-flow physical limitations and backend caveats, not with direct evidence of temporal leakage.

4. Discussion

4.1. Hydrological memory and the role of baseflow

The positive deterministic contrast E1–E0 confirms that adding a linear baseflow reservoir improves the physical RAMIS baseline, in agreement with the modelling evidence from semi-arid Andean catchments reviewed by ??: “the best simulations could be achieved with separate parameter sets for varying hydro-climatic conditions,” and

a dedicated slow-drainage component is necessary to represent the recession-dominated dry season. The NSE and RMSE improvements are modest in absolute terms but hydrologically meaningful: the baseflow reservoir introduces the delayed drainage memory that cannot be captured by an instantaneous fast-flow response alone. This conclusion echoes the quantitative demonstration of ? that including runoff memory in model inputs is “appropriate and contributes to an improvement of the models’ accuracy.”

However, the regime results in Table 4 show that low-flow simulation remains challenging despite the presence of the baseflow reservoir. A single linear store can represent average recession behaviour but is insufficient when the storage is heterogeneous across the catchment, as would be expected in an Altiplano landscape with wetlands, peatlands (bofedales) and variable permeability (?). ? showed theoretically that “between-hillslope heterogeneity alone can give rise to apparently nonlinear recession slope curves,” implying that the effective storage–discharge relationship inferred from recession analysis will always appear more complex than a simple linear reservoir. A practical consequence for StoHyMoLAP is that the baseflow improvement is bounded by the structural limitation of a single-reservoir representation. This is consistent with the finding of ? that “parallel reservoirs with independent time constants increase model performance in groundwater-dominated catchments,” suggesting a natural extension for future work.

The limited low-flow improvement also reflects a fundamental measurement challenge: the NSE criterion assigns most weight to high-flow variance (?), so the global NSE improvement from the baseflow reservoir is modest even when the absolute error in the low-flow regime is reduced. Following the recommendation of ? to use the NSE on inverse flows (NSE_{iQ}^*) to focus evaluation on the lowest 20% of flows, future studies should track this metric explicitly in addition to global NSE.

4.2. Lévy perturbations as stochastic descriptors rather than calibrated uncertainty

Lévy perturbations modify the distributional behaviour of physical simulations and, in principle, can represent the heavy-tailed variability documented for Andean precipitation and streamflow. The coverage diagnostics in Table 5, however, demonstrate that the resulting stochastic bands are strongly underdispersive: PICP values of 0.16 and 0.004 are far below the nominal 0.95, and Winkler scores reflect large penalties for missing observations. This failure to achieve calibrated coverage is not unique to Lévy-driven models: ? showed that even sophisticated multi-model ensemble systems “fail to maintain the required dispersion throughout the entire forecast horizon,” and ? found that correcting precipitation inputs alone is “obviously not enough to achieve streamflow forecast reliability.” The more general point is that calibrated uncertainty requires explicitly accounting for all sources of error simultaneously—not only stochastic forcing, but also parameter uncertainty, structural uncertainty and initial-condition uncertainty (??).

The underdispersive nature of the bands does not negate the value of Lévy perturbations within StoHyMoLAP; it merely redirects that value. Rather than serving as calibrated intervals, the stochastic ensemble summaries—particularly the quantile descriptors in Eq. 8—function as informative features that encode distributional information about the conditional response of the physical model. The key insight from ? is directly applicable here: “the spectral slopes of the rain fields play an essential role in determining the hydrological response,” and features that preserve asymmetry and tail behaviour (quantiles) carry more information than features that collapse the distribution to its mean. ? showed that quantile representations are “less sensitive to fat-tailed distributions and outliers,” a property that is particularly desirable when the ensemble members are driven by α -stable noise.

4.3. Why quantile-based hybridization works

The most robust empirical signal in this study is the advantage of quantile-based hybridization. The E6–E5 contrast adds 0.114 NSE and 0.095 KGE purely from switching the physical descriptor from a mean to a quantile vector, holding everything else constant. This result is interpretable within the framework of ?: “there are two ways for improving predictive performance—improving the prediction algorithm and discovering new informative predictors,” and quantile descriptors of the physical ensemble constitute precisely the latter. A single ensemble mean retains first-moment information but discards the conditional spread, asymmetry and tail behaviour of the physical simulation. In the Ramis basin, where the flow regime alternates sharply between wet-season peaks and dry-season recession, these distributional properties are not constant: the upper quantiles expand during the wet season while the lower quantiles collapse. This asymmetric dynamics is informationally invisible to a mean-only descriptor and yet is directly observable in the quantile vector.

The hybrid emulator (E6) also outperforms the pure ML baseline (E4) despite E4 having access to observed meteorological forcing. This indicates that physically simulated ensemble descriptors provide information beyond what is contained in the observed forcings alone, a result consistent with ? who found NSE improvements of 23%

when ML surrogates were enriched with synthetic physical data, and with ? who showed that SWAT outputs “provide a proxy about the instantaneous wetness condition of the catchment.” The interpretation is that the physical ensemble descriptors effectively encode catchment antecedent state in a form that the ML emulator can more readily exploit than raw meteorological time series. This is consistent with the argument of ? that “a machine learning model needs a teacher (hydrologist) that provides it with textbooks and then leaves it for self-teaching,” and with the finding of ? that models lacking antecedent runoff information cannot compete with physically constrained alternatives.

One should note, however, that the E6 advantage over E4 is partly confounded with the difference in feature type (ensemble statistics versus raw forcing), and the E6–E4 contrast alone cannot distinguish between the value of hydrological structure and the value of temporal aggregation. Resolving this would require an additional ablation experiment pairing observed forcing with quantile features derived from simple lag statistics, which is left for future work. This caveat is consistent with the general warning of ? that “implicit coupling... requires more elaborate design of the model structure and inputs, as well as more thorough verification of the results to avoid learning other misleading relationships.”

4.4. Backend-aware interpretation of the sequential experiments

E8 is the top-ranked configuration in Table 2, but the result must be interpreted together with Table 6. The fallback MLP backend in E7 and E8 processes flattened sequences rather than recurrent hidden states, and therefore does not implement the temporal gating mechanism that is the defining feature of GRU cells (??). The valid conclusion is narrower: a GRU-labelled sequential hybrid using quantile descriptors and implemented with a fallback flattened-sequence MLP achieved the best validation metrics in the current environment. Whether a true recurrent GRU would reproduce or exceed this performance cannot be established from the current evidence package.

The consistent ordering of quantile over mean descriptors in both the XGB-labelled (E6>E5) and the GRU-labelled (E8>E7) families suggests that the descriptor choice is the driver of improvement rather than the emulator architecture. This is an important practical conclusion: it implies that the gain from quantile hybridization should be architecture-agnostic, and therefore reproducible under a true GRU backend. Future work should rerun E7 and E8 with TensorFlow or PyTorch backends and compare the GRU-specific gain to the descriptor-specific gain measured here. Transparent backend reporting is essential for such comparison, as emphasized by ?: “open hydrologists document all components of their data collection and analysis pipeline, favoring open and non-proprietary technologies.”

4.5. Leakage, reproducibility and submission readiness

The audit status in Table 7 strengthens the comparison by documenting the major leakage controls: strict temporal splitting, non-negative lags, sequence construction within temporal subsets, train-only scalers and train-only regime thresholds. “Only 1% of hydrology papers were fully reproducible” in a recent large-scale survey (?), and the StoHyMoLAP audit is an explicit step toward closing that gap.

The WARN status should remain visible in the manuscript or supplementary material. WARN does not indicate detected temporal leakage, but it does identify issues that affect interpretation: missing observed-flow values, physical low-flow limitations, underdispersive stochastic bands and fallback ML backends. These are honest disclosures consistent with the open-science principles advocated by ? and with the rigorous benchmarking framework recommended by ?.

5. Limitations

This study has five main limitations. First, the sequential experiments labelled as GRU were not executed with a real recurrent backend, so their results cannot support claims about recurrent neural networks (?). Second, the Lévy stochastic bands are underdispersive and should not be treated as calibrated uncertainty intervals; recalibration following the guidelines of ? and ? is required before these bands can be used operationally. Third, low-flow bias remains substantial despite improvements in absolute-error metrics; future work should apply the inverse-flow NSE criterion of ? to focus calibration explicitly on the dry-season regime. Fourth, the audit status is WARN rather than fully PASS, requiring transparent reporting of missing observations and physical-audit warnings. Fifth, confirm data provenance and units (Section 2.1): if the current package corresponds to an illustrative or reduced-computation run, or if the discharge series is normalized or refers to a sub-catchment rather than the full Puente Ramis basin, the final submission must regenerate the complete evidence package with the definitive, traceable observed Ramis dataset (source, station, units) and publication-level calibration settings.

6. Conclusions

StoHyMoLAP provides a reproducible, audit-controlled framework for evaluating stochastic physical–machine-learning hybridization in daily streamflow simulation, applied here to the high-Andean Ramis basin in the Lake Titicaca system. Four main conclusions emerge from the controlled ablation.

First, quantile descriptors from the stochastic physical ensemble are the dominant driver of hybrid improvement: replacing the ensemble mean with quantile features added 0.114 NSE (E6–E5), and the full quantile hybrid surpassed the physical model by 0.311 NSE (E6–E3). This confirms, in an Altiplano setting, the argument of ? and the quantitative demonstration of ? that distributional descriptors carry information beyond a single trajectory.

Second, the baseflow reservoir improves the deterministic physical baseline (E1–E0: $\Delta\text{NSE} = 0.024$) by introducing slow-drainage memory consistent with the groundwater-dominated response of the Ramis system (?), but a single linear reservoir is insufficient to fully correct the low-flow bias documented by ? and ?.

Third, Lévy perturbations improve point prediction inconsistently as a direct physical forcing (E3–E1: $\Delta\text{NSE} = -0.010$) but contribute positively as generators of ensemble diversity whose distributional summaries serve as informative ML features. The stochastic bands are strongly underdispersive (PICP = 0.004 for E3) and cannot be used as calibrated predictive intervals without further post-processing (??).

Fourth, the numerically top-ranked configuration (E8, $\text{NSE} = 0.849$) must be reported as a GRU-labelled fallback-MLP run rather than as evidence of recurrent neural-network superiority (?). The descriptor ordering (quantile > mean) is nevertheless consistent across both emulator families, supporting the conclusion that the information value of distributional physical descriptors is architecture-agnostic.

Future work should regenerate E7 and E8 with real recurrent backends, recalibrate stochastic uncertainty toward nominal coverage, apply inverse-flow NSE criteria to improve low-flow performance, and evaluate the framework at full-resolution Ramis data before transferring to neighbouring Altiplano catchments.

Code and data availability

The manuscript tables and figures are generated from the StoHyMoLAP Phase 3.4B–3.5 result package. The scripts `make_method_diagram.py` and `build_paper_assets.py` reproduce all figures and CSV tables. Backend metadata should be regenerated after any change to the ML environment. repository URL/DOI and the SENAMHI/ANA data-source citation.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this draft, AI-assisted writing was used to organize the manuscript, convert reproducible artefacts into narrative text, and generate supporting LaTeX scripts. The authors remain responsible for verifying all scientific claims, numerical values, references, data provenance and final wording before submission.

CRedit authorship contribution statement

Jose Zevallos: Conceptualization of this study, Methodology, Software, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

Jose Zevallos: Conceptualization of this study, Methodology, Software, Writing – original draft.

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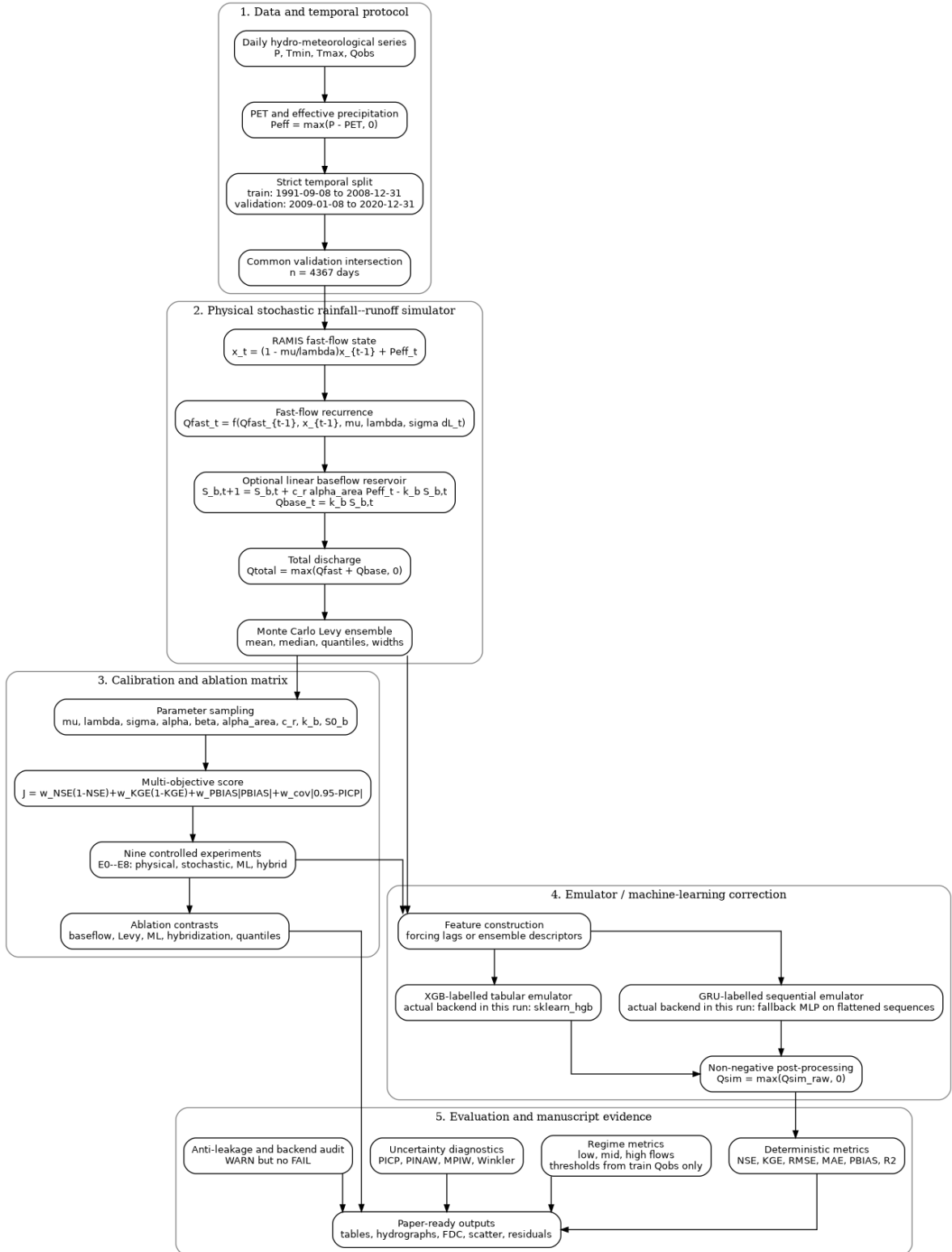


Figure 1: Hierarchical workflow of the StoHyMoLAP methodology, from daily hydro-meteorological data to physical simulation, stochastic ensemble generation, emulator training, ablation comparison and audit-controlled manuscript evidence.

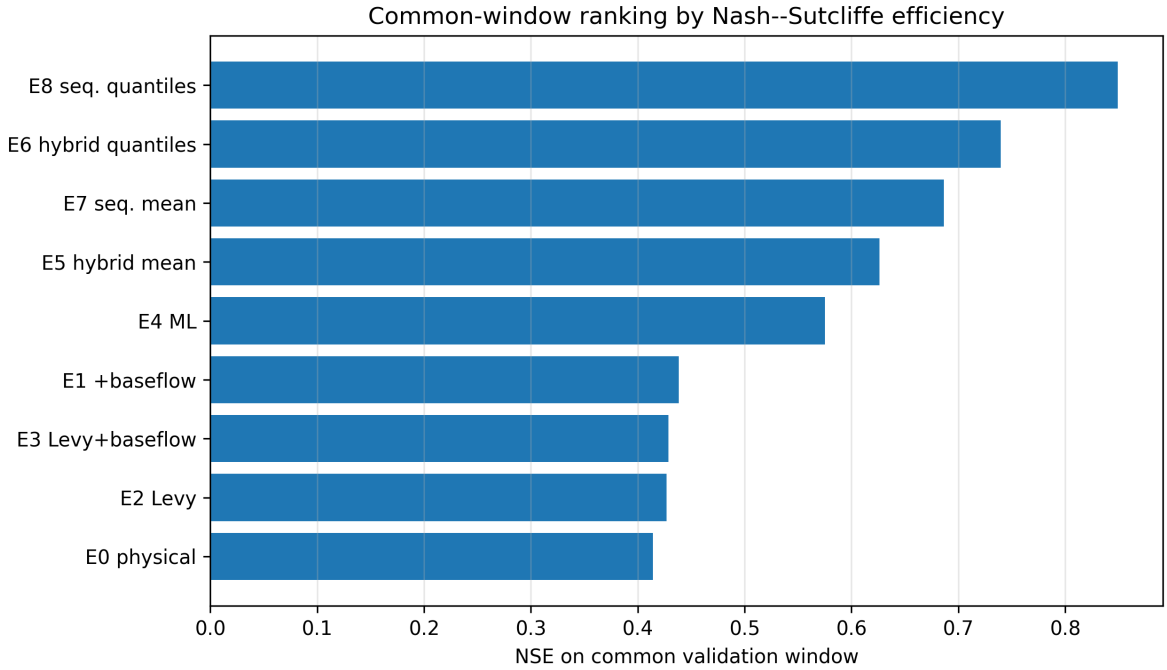


Figure 2: Ranking of all experiments by common-window NSE. Numerical values correspond to Table 2.

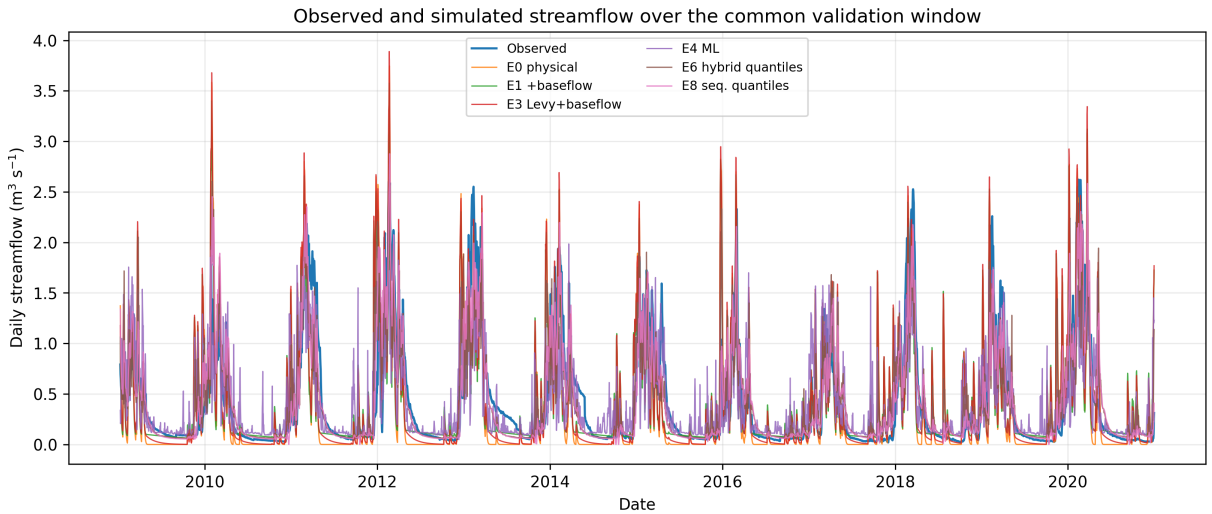


Figure 3: Observed and simulated daily discharge over the common validation window for representative physical, pure ML and hybrid models.

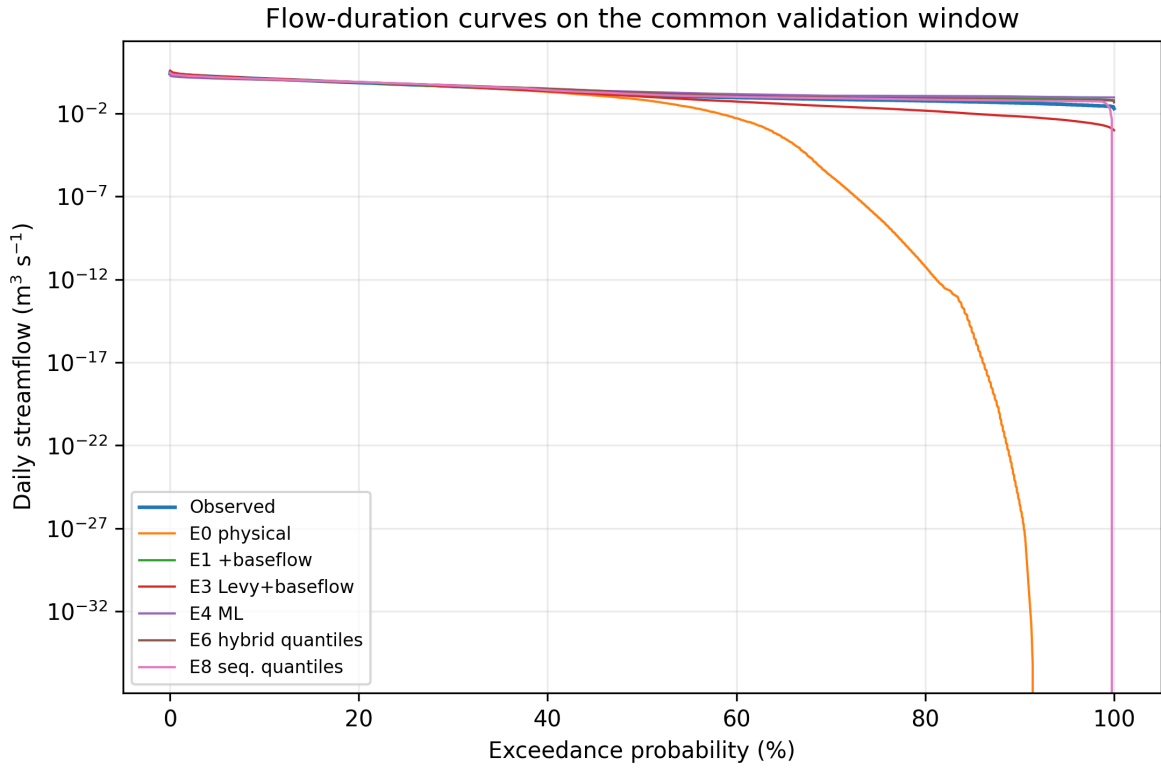


Figure 4: Flow-duration curves for observed discharge and representative simulations over the common validation window.

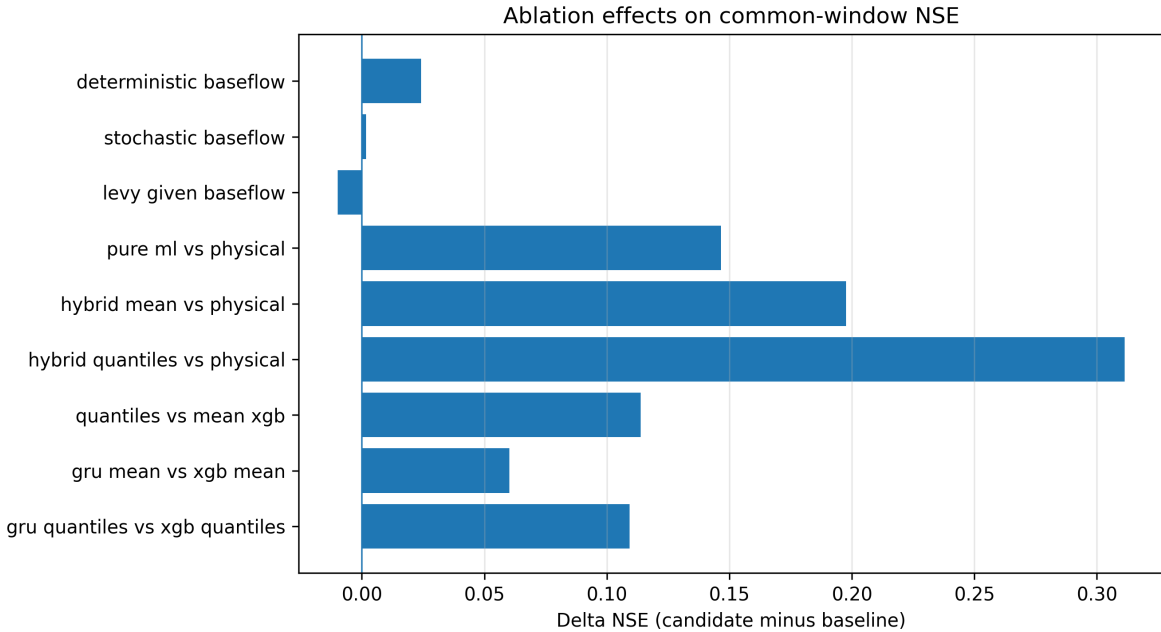


Figure 5: Ablation effects expressed as candidate-minus-baseline Δ NSE. Numerical values are reported in Table 3.

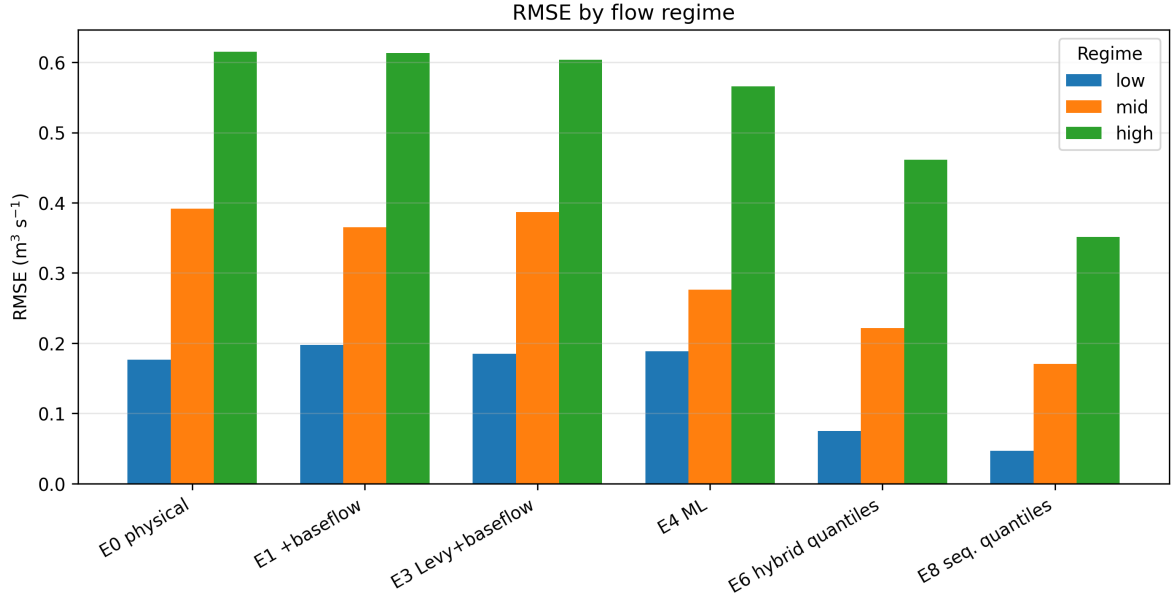


Figure 6: RMSE by low, mid and high flow regimes for representative models. Values correspond to Table 4.

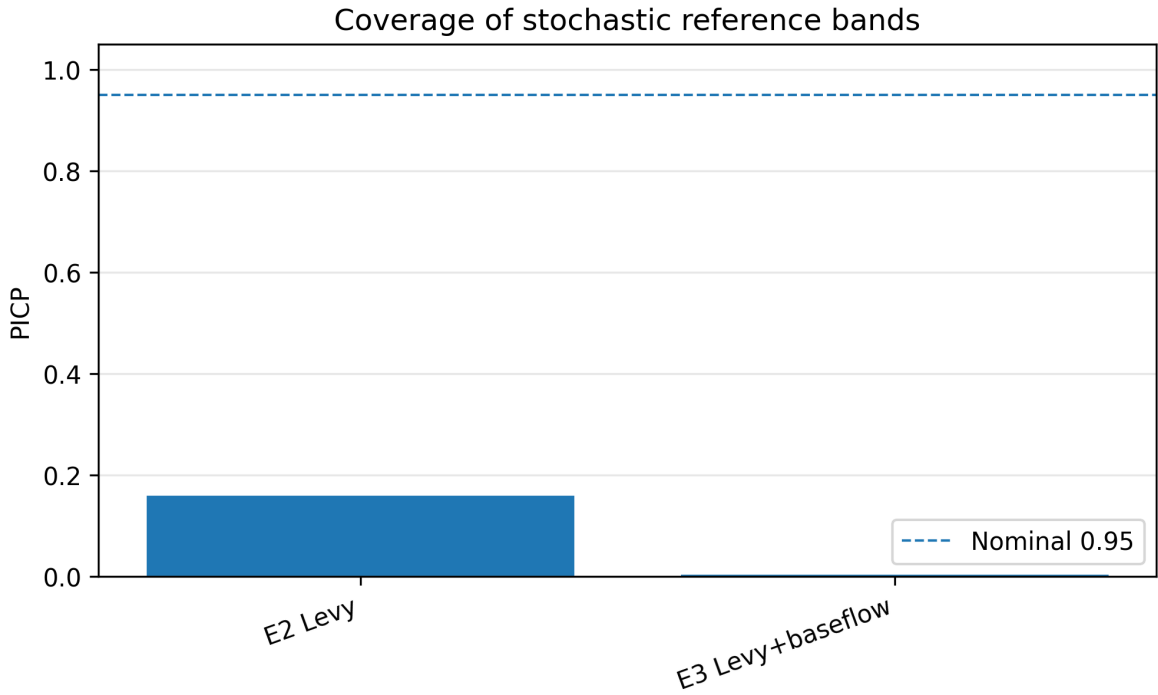


Figure 7: Prediction interval coverage probability for stochastic physical references. The dashed line marks the nominal 0.95 target.